Inferring Audience Interests from Comment Streams with Transformer Sentence Embeddings: a Kernel Mean Embedding View of Ad Targeting

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Abstract

We formalise the problem of inferring the interests of the audience of a social-media channel from the public comments it receives, in order to match advertisers with audiences rather than with content. Comments are mapped by a contrastively fine-tuned BERT encoder to the unit sphere; a channel is then represented by the *kernel mean embedding* of its comment distribution. We show that (i) under a topic-mixture model of the audience, this embedding is an affine image of the average interest vector and is identifiable up to a linear map, (ii) its weighted empirical estimator concentrates at rate $\|\boldsymbol{w}\|_2$, (iii) inner products with an advertiser query equal the expected query affinity of a random commenter, which, under the InfoNCE optimum, approximates a pointwise mutual information. Look-alike targeting reduces to a maximum mean discrepancy. On 3 154 French YouTube channels, ranking by these statistics recovers channels already sponsored by a given advertiser with AUC 0.87.

1. Problem

Let $\mathcal{C}$ be a set of channels and $\mathcal{X}$ the space of finite token strings. The audience of $c \in \mathcal{C}$ is a population U_c of users, each endowed with a latent interest vector $\theta_u \in \Delta^{K-1}$ over K interest topics. We observe only comments $X_1, \dots, X_n \in \mathcal{X}$ written by members of U_c. An advertiser describes a target by a text q. The goal is a score $s(c, q)$ that is monotone in the expected interest of U_c for q, computable from comments alone, and that does not require any individual-level profile.

2. BERT and sentence embeddings

Encoder. A string is tokenised into $t_1, \dots, t_L$ and embedded as $H^{(0)} = E_{\text{tok}} + E_{\text{pos}} \in \mathbb{R}^{L \times d}$. Each of the N layers of a Transformer encoder [16] applies multi-head self-attention,

$$\mathrm{Att}(Q, K, V) = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_h}} + B\Big)V, \quad Q = HW_j^Q,\ K = HW_j^K,\ V = HW_j^V, \tag{1}$$

(B masks padding), concatenates the h heads, and composes with residual connections, layer normalisation and a position-wise network: $H' = \mathrm{LN}(H + \mathrm{MHA}(H))$, $H^+ = \mathrm{LN}(H' + W_2\,\mathrm{GELU}(W_1 H'))$. Every output vector $h_i^{(N)}$ is thus a context-dependent function of the whole sequence. BERT [3] is pre-trained by masked language modelling: for a random subset M ($\approx 15\%$) of positions,

$$\mathcal{L}_{\mathrm{MLM}}(\vartheta) = -\mathbb{E}_{x,M} \sum_{i \in M} \log p_\vartheta\big(t_i \mid x_{\setminus M}\big), \tag{2}$$

which forces $h_i^{(N)}$ to encode the distributional semantics of the context.

Sentence map. Token states are mean-pooled and normalised, $\phi(x) = \bar{h}/\|\bar{h}\|$, $\bar{h} = \frac{1}{L}\sum_i h_i^{(N)}$, so $\phi : \mathcal{X} \to \mathbb{S}^{d-1}$. Raw MLM states are strongly anisotropic [4]; Sentence-BERT [13, 14] and E5 [17] therefore fine-tune ϕ on pairs (x, x^+) with the InfoNCE loss [11]

$$\mathcal{L} = -\mathbb{E} \log \frac{\exp(\langle \phi(x), \phi(x^+)\rangle/\tau)}{\sum_{j=0}^{m} \exp(\langle \phi(x), \phi(x_j)\rangle/\tau)}, \tag{3}$$

with $x_0 = x^+$ and m negatives. At the population optimum of an unconstrained critic f, $\exp f(x, y) \propto p(y \mid x)/p(y)$ [11]; hence, to the extent the encoder attains it,

$$\langle \phi(x), \phi(y)\rangle \approx \tau\,\mathrm{PMI}(x; y) + \mathrm{const}(x). \tag{4}$$

Cosine similarity is thus a (scaled) estimate of semantic association, and the bilinear form $k(x, y) = \langle \phi(x), \phi(y)\rangle$ is a bounded positive-definite kernel on $\mathcal{X}$ with $k(x, x) = 1$.

3. Audience as a distribution

Definition 1. *Let P_c be the law of a comment written by a random member of U_c. The* audience embedding *of c is the kernel mean embedding [15, 10] $\mu_c = \mathbb{E}_{X \sim P_c}[\phi(X)] \in \mathbb{B}^d$.*

Since ϕ is bounded, μ_c exists, and by linearity, for any query embedding $\psi(q) = \phi(q)$,

$$\langle \mu_c, \psi(q)\rangle = \mathbb{E}_{X \sim P_c}\, k(X, q), \tag{5}$$

i.e. the expected affinity of a random commenter with the advertiser text; by (4) it is $\tau\,\mathbb{E}_{P_c}\mathrm{PMI}(X; q)$ up to an additive term.

Topic mixture. Assume, as in LDA [1], that a user with interests θ writes $X \sim \sum_k \theta_k Q_k$, Q_k the comment law of topic k, and let ρ_c be the law of θ_u on U_c.

Proposition 1 (linear identifiability)**.** *Let $m_k = \mathbb{E}_{Q_k}\phi$, $M = [m_1 | \cdots | m_K]$ and $\bar{\pi}_c = \mathbb{E}_{\rho_c}\theta$. Then $\mu_c = M\bar{\pi}_c$. If $\mathrm{rank}\, M = K$, $\bar{\pi}_c = M^+\mu_c$ is uniquely determined by μ_c, and for every query $\langle \mu_c, \psi(q)\rangle = \sum_k \bar{\pi}_{c,k}\, a_k(q)$, $a_k(q) = \langle m_k, \psi(q)\rangle$.*

Proof. By Fubini, $\mu_c = \int \sum_k \theta_k m_k \, d\rho_c(\theta) = M\bar{\pi}_c$; full column rank gives $M^+M = I_K$. □

The score therefore ranks channels by the *average interest of their audience*, weighted by how strongly each topic relates to the product. Only $\bar{\pi}_c$—not the higher moments of ρ_c—is identifiable from μ_c with a linear kernel; heterogeneity is recovered by the sub-audience decomposition below.

Common component and hubness. All comment laws share a generic component (greetings, "great video"), so $m_k = g + \tilde{m}_k$ with a large common g and cosines are tightly concentrated. We therefore use the centred embedding $\mu_c - g$, $g = |\mathcal{C}|^{-1}\sum_c \mu_c$, and correct hubness [12] as in CSLS [2]: with a set $\mathcal{P}$ of unrelated probe topics,

$$\tilde{s}(c, q) = \langle \mu_c, \psi(q)\rangle - \tfrac{1}{|\mathcal{P}|}\sum_{p \in \mathcal{P}} \langle \mu_c, \psi(p)\rangle = \langle \mu_c, \psi(q) - \bar{\psi}_{\mathcal{P}}\rangle, \tag{6}$$

still linear in μ_c, so Prop. 1 applies with $a_k(q)$ replaced by its excess over a random topic. Scores are finally standardised across channels (z-scores), since only the relative order matters for targeting.

4. Estimation

Comments are not i.i.d. across videos and their likes encode endorsement. We take deterministic weights $w_i \geq 0$, $\sum_i w_i = 1$, giving each video equal mass split proportionally to $1 + \log(1 + \text{likes}_i)$, and set $\hat{\mu}_c = \sum_i w_i \phi(X_i)$.

Proposition 2 (concentration)**.** *If* $X_1, \dots, X_n \overset{\text{iid}}{\sim} P_c$, *then with probability at least* $1 - \delta$,

$$\|\hat{\mu}_c - \mu_c\| \leq \|\boldsymbol{w}\|_2 \big(1 + \sqrt{2\log(1/\delta)}\big),$$

and $|\langle \hat{\mu}_c - \mu_c, \psi(q)\rangle|$ *obeys the same bound uniformly in* q.

Proof. $\mathbb{E}\|\hat{\mu}_c - \mu_c\|^2 = \sum_i w_i^2 \mathbb{E}\|\phi(X_i) - \mu_c\|^2 \leq \|\boldsymbol{w}\|_2^2$, and Jensen bounds the mean of the norm. Replacing X_i changes $\|\hat{\mu}_c - \mu_c\|$ by at most $2w_i$; McDiarmid's inequality [9] gives $\Pr(\cdot - \mathbb{E}\,\cdot \geq t) \leq \exp(-t^2/(2\|\boldsymbol{w}\|_2^2))$. Cauchy–Schwarz with $\|\psi(q)\| = 1$ gives the second claim. □

The effective sample size is $n_{\text{eff}} = \|\boldsymbol{w}\|_2^{-2} \leq n$; the bound is dimension-free, which is what makes $d = 384$ embeddings usable with a few hundred comments per channel.

Sub-audiences. Weighted k-means [8] minimises $\sum_i w_i \|\phi(X_i) - \gamma_{\kappa(i)}\|^2$ and yields centroids $\gamma_1, \dots, \gamma_k$ with masses ω_j; the score $\max_{j:\omega_j \geq 0.1} \langle \gamma_j, \psi(q)\rangle$ detects a niche segment that the mean dilutes.

5. From scores to targeting

Response model. Suppose a user who writes x responds to the ad with probability $r(x) = \alpha + \beta\, k(x, q)$, $\beta > 0$ (first-order expansion of any smooth link). Then the expected response rate of U_c is $\alpha + \beta\langle \mu_c, \psi(q)\rangle$, so ranking by (5) is *exactly* ranking by expected conversion; for a nonlinear link σ, the error is a Jensen gap of order $\frac{1}{2}\sup|\sigma''|\,\mathrm{Var}_{P_c} k(X, q)$. With reach v_c (expected views) and price π_c, the advertiser solves a knapsack $\max \sum_c y_c\, v_c(\alpha + \beta\tilde{s}_c)$ s.t. $\sum_c y_c \pi_c \leq$ budget.

Facets. A target "travellers, price-sensitive, worried about public Wi-Fi" encoded as one vector averages its parts. We score each facet q_f separately, $T_f(c)$, and combine $S_\lambda(c) = \overline{T}(c) - \lambda\,\mathrm{sd}_f\, T_f(c)$. For $F = 2$, $\overline{T} - \mathrm{sd}\, T = \min(T_1, T_2)$, so $\lambda \in [0, 1]$ interpolates between the mean and a logical AND.

Look-alike. Given channels R already sponsored by a competitor, let $\nu_R = |R|^{-1} \sum_{r \in R} (\mu_r - g)$ and score $\cos(\mu_c - g, \nu_R)$. Since $\|\mu_c - \mu_r\|^2 = \mathrm{MMD}_k^2(P_c, P_r)$ [5], this is a nearest-neighbour rule in the maximum mean discrepancy between audiences, independent of any textual description of the product.

6. Empirical evidence and limits

On 3 154 French channels (`multilingual-e5-small`, $d = 384$), with ground truth the 102 channels sponsored by a VPN brand (base rate 3%), the area under the ROC curve, $\mathrm{AUC} = \Pr(s(C^+) > s(C^-))$ [6], is 0.68–0.76 for text queries and 0.87 for look-alike, with an 18× lift in precision at 20.

Three limits follow from the model. (a) P_c is the law of *commenters*, a selected subset of viewers; without propensities the estimate is biased toward engaged users. (b) Comments react to video content, so μ_c mixes audience and content; subtracting a content embedding isolates "hidden" audiences whose interests the creator does not discuss. (c) Demographic attributes are not identifiable this way: they do not enter $k(X, q)$ through semantics but through lexical cues. Finally, although private traits are predictable from digital traces [7], the method only ever estimates population means μ_c, in line with data minimisation.

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